Unit 3

Machine Learning

Types of Evaluation Metrics Accuracy, Precision, Recall or F1?

Example: Predicting Diabetes (+ve / −ve)

Let’s consider a scenario with **100 individuals**.

- **Actual (True Labels)**
  - 45 people **have diabetes** (+ve)
  - 55 people **do not have diabetes** (−ve)

A machine learning model predicts whether a person has diabetes based on medical data.

Model Predictions

Category	Count
Predicted **Diabetes (+ve)**	50
Predicted **No Diabetes (−ve)**	50

True Positive (TP)

- People **who actually have diabetes**
- Model **correctly predicts diabetes**

Out of 45 diabetic patients, **40** were correctly predicted.

TP = 40

False Negative (FN)

- People **who actually have diabetes**
- Model **predicts no diabetes**

Remaining diabetic patients = 45 − 40 = **5**

FN = 5

False Positive (FP)

- People **who do NOT have diabetes**
- Model **incorrectly predicts diabetes**

Out of 55 non-diabetic people, **10** were wrongly predicted as diabetic.

FP = 10

True Negative (TN)

- People **who do NOT have diabetes**
- Model **correctly predicts no diabetes**

Remaining non-diabetic people = 55 − 10 = **45**

TN = 45

Evaluation Matrix (Confusion Matrix)

An **evaluation matrix** (confusion matrix) shows how well a classification model performs by comparing **actual values** with **predicted values**.

Binary Classification Confusion Matrix

	Predicted Positive	**Predicted Negative**
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

	Predicted Positive	**Predicted Negative**
Actual Positive	40	5
Actual Negative	10	45

Meaning:

- **TP**: Correctly predicted positive
- **TN**: Correctly predicted negative
- **FP**: Incorrectly predicted positive
- **FN**: Incorrectly predicted negative

Accuracy

How many predictions were correct overall?

Formula:

$$\textbf{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Example:

$$\text{Accuracy} = \frac{40 + 45}{40 + 45 + 10 + 5}$$

$$\text{Accuracy} = \frac{85}{100}$$

$$\boxed{\text{Accuracy} = 0.85 \ = \ 85\%}$$

Precision

Out of all predicted positives, how many are actually positive?

Formula:

$$\textbf{Precision} = \frac{TP}{TP + FP}$$

Example:

$$\text{Precision} = \frac{40}{40 + 10}$$

$$\text{Precision} = \frac{40}{50}$$

$$\boxed{\text{Precision} = 0.80 \ = \ 80\%}$$

Important when **false positives are costly**
Example: Spam detection

Recall (Sensitivity)

Out of all actual positives, how many were correctly predicted?

Formula:

$$\textbf{Recall} = \frac{TP}{TP + FN}$$

Example:

$$\text{Recall} = \frac{40}{40 + 5}$$

$$\text{Recall} = \frac{40}{45}$$

$$\boxed{\text{Recall} \approx 0.89 \ = \ 89\%}$$

Important when **false negatives are dangerous**
Example: Disease detection

F1-Score

Harmonic mean of Precision and Recall

Formula:

$$\mathbf{F1\text{-}score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Example:

$$\text{F1} = \frac{2 \times (0.80 \times 0.89)}{0.80 + 0.89}$$

$$\text{F1} = \frac{2 \times 0.712}{1.69}$$

$$\boxed{\text{F1-score} \approx 0.84 \;=\; 84\%}$$

Best when **data is imbalanced**